

Embedded System 9EE-53EA

Scheme of Evaluation

SECTION- I

- | | |
|---------------------------|------------|
| 1(a) Fill up the blanks : | 1x5=5marks |
| 1(b) Any five comparisons | 1x5=5marks |

SECTION- II

- | | |
|--|--------------------|
| 2(a) Any five features | 1x5=5 |
| 2(b) Block Diagram | 5marks |
| 2(c) Explanation | 2.5x2 =5 marks |
| 3(a) Listing +example for each | (5x0.5)+(5x0.5) =5 |
| 3(b)Explanation : Timer0 – 2.5 Marks Timer1 – 2 .5Marks | 2.5x 2=5 |
| 3(c) Diagram + function of 8 bits | 1+(8x0.5) =5 marks |
| 4(a) (i) binary addition :1 mark
Status of CY:1mark
Status of AC:1 mark
ii) Binary addition 1 mark
Status of CY:1mark
Status of AC:1 mark | 2x 3 =6 marks |
| 4(b) Correct explanation | 2x3=6marks |
| 4(c) Correct explanation | 3 |

SECTION- III

- | | |
|---|-----------|
| 5(a) Clearing Accumulator – 1.5Mark
Initialising Counter – 1.5 Mark
Loop to add 3 to accumulator 10 times – 3 Mark | 6 |
| 5(b) Alternate function of each pin of Port 3 – 0.5 Marks | 0.5 x 8=4 |
| 5(c) correct program
Loading data from RAM locations to Register A and B=1 mark
Using of MUL instruction=2 mark
Storing result in RAM locations= 1 mark
Comments= 1 marks | 5 marks |

6(a) Correct program 7 marks

Main Program : TMOD instruction for mode selection 1 Mark

Loading TL0 and TH0 registers for time delay 1 Mark

Compliment bit instruction 1 Mark

Calling delay routine 1 Mark

Delay Program: Timer start instruction 1 Mark

Monitoring Timer flag until it rolls over 1 Marks

Stop timer and clearing timer 0 flag 1 Mark

6(b) figure+explanation 4+4=8marks

7(a) Finding TH1 value to get baud rate 9600 2.5 Marks

Finding TH1 value to get baud rate 4800 2.5 Marks

5 marks

7(b) Listing Interrupts Highest to lowest priority

5 marks

7(c) explanation

5 marks

SECTION- IV

8(a) pin description 5marks

8(b) Diagram 5 marks

8(c) Correct C program 5 marks

correct Main function=3 marks

correct Delay function= 2 marks

9(a) Diagram 5 marks

9(b) Explanation 5 marks

9(c) Correct C program 5 marks

Assigning of RD,INTR and WR pins to 8051 pins=1

Main function=4marks

10(a) Diagram 4 marks

10(b) Diagram 6 marks

10(c) Diagram 5 marks

Certified that Model answers prepared by me for subject code No. 9EE53EA are from prescribed textbook and model answers and scheme of valuation, prepared by me are correct and unambiguous

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Embedded System 9EE-53EA

MODEL ANSWERS

SECTION-I

1(a)

- i. 8 bit
- ii. TCON
- iii. Serial Control
- iv. Asynchronous
- v. 10mV

1(b) Compare microprocessor with Microcontroller.

5

Sl no	Microprocessor	Microcontroller
1	Microprocessor is the heart of Computer system.	Micro Controller is the heart of an embedded system.
2	Cost of the entire system is high	Cost of the entire system is low
3	<i>The total consumption of power is HIGH</i>	<i>The total consumption of power is LOW</i>
4	It has no RAM, ROM, Input-Output units, timers, and other peripherals on the chip	, It has a CPU along with RAM, ROM, and other peripherals embedded on a single chip.
5	It's used for general purpose applications that allow you to handle loads of data	It's used for application-specific systems.
6	Microprocessor has a smaller number of registers, so more operations are memorybased.	Microcontroller has more register. Hence the programs are easier to write

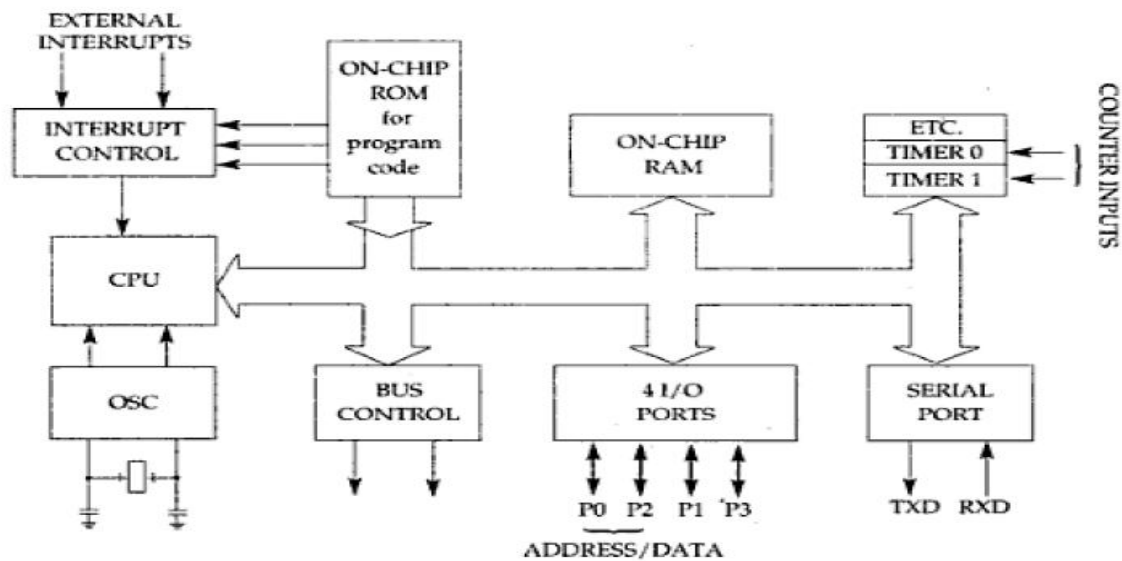
SECTION-II

2(a)

Any five Features of microcontroller

5x1=5

- i. 128 bytes of internal RAM
- ii. 4K bytes of ON chip ROM
- iii. Two 16 bit timers
- iv. Four 8-bit ports(P0-P3)
- v. One full duplex serial port
- vi. 16-bit program counter(PC) and 16-bit data pointer(DPTR)
- vii. Six interrupts including reset
- viii. 8-bit CPU with Register A and B
- ix. 8-bit stack pointer and 8-bit PSW



2(c) Explain program counter and Data pointer

(2.5x2)

Program counter(PC) :

- Program counter is a 16 bit register and it contains 16 bit address
- PC Points to the address of the next instruction to be executed
- As each instruction is executed, the program counter is incremented to point to the address of the next instruction to be executed
- PC can have address range from 0000h to FFFFh
- PC is the only register that does not have any address

Data pointer :

- It is a 16 bit register
- It consists of two 8-bit registers (DPH and DPL)
- They are used to access external data
- They are used to access internal and external code access
- DPTR is under the control of program instructions

3(a) List five addressing modes of 8051 and example for each

(2.5+2.5)

The five addressing modes of 8051 with example for each are :

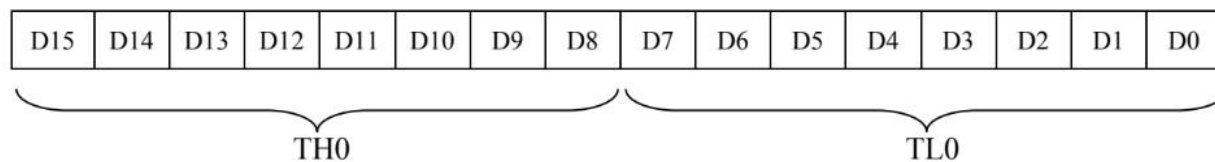
- Immediate addressing : `MOV A, #20h`
- Register addressing : `MOV A, R1`
- Direct addressing : `MOV A, 20h`
- Register indirect : `MOV A, @R1`

v. Indexed addressing : `MOVC A, @A+DPTR`

3(b) Explain Timer registers of 8051

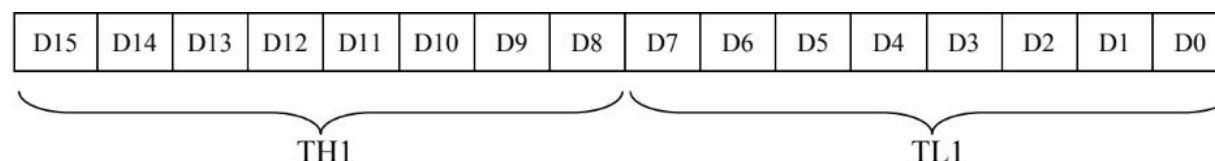
The 8051 has two timers: Timer0 and Timer1. Both are 16-bits wide. Since 8051 has an 8-bit architecture, each 16-bit timer is accessed as two separate registers of low byte and high byte.

Timer0 register:



The 16-bit register of timer0 is accessed as low byte register called as TL0 and high byte register called TH0. These registers can be accessed like any other register.

Timer1 register:



Timer1 is also 16 bits and it is split into two bytes, called as TL1(Timer1 low byte) and TH1(Timer1High byte). These registers are accessible in the same way as Timer0 registers.

3(C) Draw IE sfr. Explain function of each bit . (5)

IE register is used to enable and disable interrupts. fig. shows IE register and function of each bit

EA	---	ET2	ES	ET1	EX1	ET0	EX0
----	-----	-----	----	-----	-----	-----	-----

EA (IE .7): EA=0 disable all interrupts; EA=1 permit individual interrupt to be enabled or disabled.

--- (IE .6): Not implemented

ET2 (IE .5): Reserved for future use

ES (IE .4): ES=1 enable serial interrupt; ES=0 disable serial interrupt

ET1 (IE.3): ET1=1 enable timer1 interrupt; ET1=0 disable timer1 interrupt

EX1 (IE.2): EX1=1 enable external interrupt 1; EX1=0 disable external interrupt 1

ET0 (IE.1): ET0=1 enable timer 0 interrupt; ET0=0 disable timer 0 interrupt

EX0 (IE.0): EX0=1 enable external interrupt 0; EX0=0 disable external interrupt 0

4(a) what is the status of CY and AC flags after addition in each of the following cases:

3 + 3

(i) MOV A, #38H	38	0011 1000
ADD A, #2FH	<u>+2F</u>	<u>0010 1111</u>
	67	0110 0111

CY = 0 since there is no carry beyond the D7 bit.

AC = 1 since there is a carry from D3 to D4 bit.

(i) MOV A, #9CH	9C	1001 1100
ADD A, #64H	<u>+64</u>	<u>0110 1000</u>
	100	10000 0100

CY = 1 since there is a carry beyond the D7 bit.

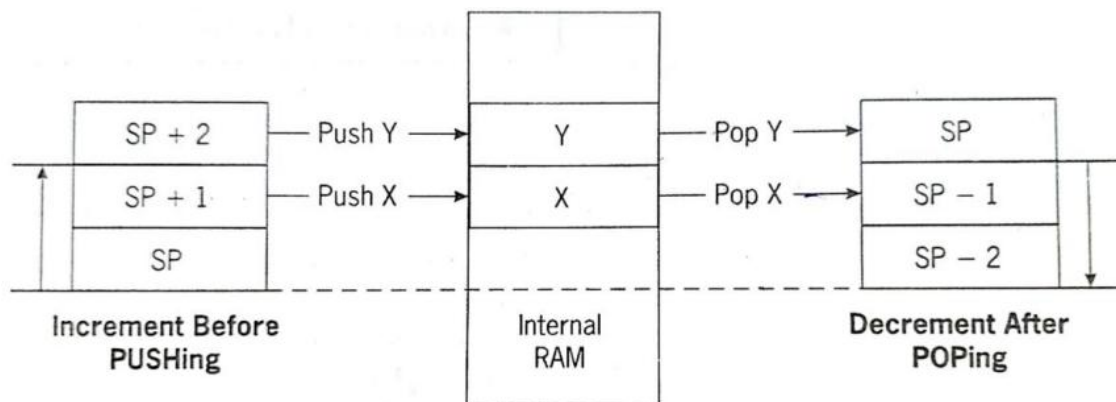
AC = 1 since there is a carry from D3 to D4 bit.

4(b) Explain PUSH and POP operation

3 + 3

PUSH opcode copies data from the source address to the stack. SP is incremented by 1 before the data is copied to the internal RAM location contained in SP so that the data is stored from low addresses to high addresses in the internal RAM. The stack grows up in memory as it is PUSHed. Excessive PUSHing can make the stack exceed 7Fh (the top of internal RAM), after which point data is lost.

POP opcode copies data from the stack to the destination address. SP is decremented by 1 after data is copied from the stack RAM address to the direct destination to ensure that data placed on the stack is retrieved in the same way as it was stored. This is explained in the below shown diagram.



4(c) Explain the instructions XRL

3

Mnemonic: XRL <dbyte>, <sbyte>

Operation: This instruction performs the logical XOR on the two operands and places the result in the destination variable.

Example:

MOV A, #47h ; A = 47h

XRL A, #35h ; A = 47h XOR 35h = 72h

47h = 0100 0111

35h = 0011 0101

XOR 0111 0010

Hence after the above XRL instruction, A = 72h.

SECTION-III

5 (a) Write assembly program to add 3 to accumulator 10 times.

6

MOV A, #0

MOV R2, #10

AGAIN : ADD A, #03

DJNZ R2, AGAIN

MOV R5, A5

5(b) List alternate functions of Port 3 pins

4

P3 bit	Function	Pin No.
P3.0	RxD	10
P3.1	TxD	11
P3.2	$\overline{INT0}$	12
P3.3	$\overline{INT1}$	13
P3.4	T0	14
P3.5	T1	15
P3.6	$\overline{WR}$	16
P3.7	$\overline{RD}$	17

5(c) PROGRAM TO MULTIPLY TWO NUMBERS

(5 marks)

MOV A, 35H ; load contents of RAM location to accumulator

MOV B, 36H ; load contents of RAM location to Register B

MUL AB ; multiply contents of A and B

MOV 37H, A ; store the lower byte of result in register A

MOV 38H, B ; store the Higher byte of result in register B

END

6(a) Write an 8051 program to generate square wave of 50% duty cycle

7

Note: program can be written either in assembly language or C language.

```
MOV TMOD, #01    ; Timer 0, mode 1

HERE: MOV TL0, #0F2H    ; TL0=F2H, the low byte

      MOV TH0, #0FFH    ; TF0=FFH, the high byte

      CPL P1.5          ; Toggle P1.5

      ACALL DELAY        ; Call delay subroutine

      SJMP HERE         ; Load TH, TL again

;-----DELAY USING TIMER 0

DELAY: SETB TR0    ; Start the timer 0

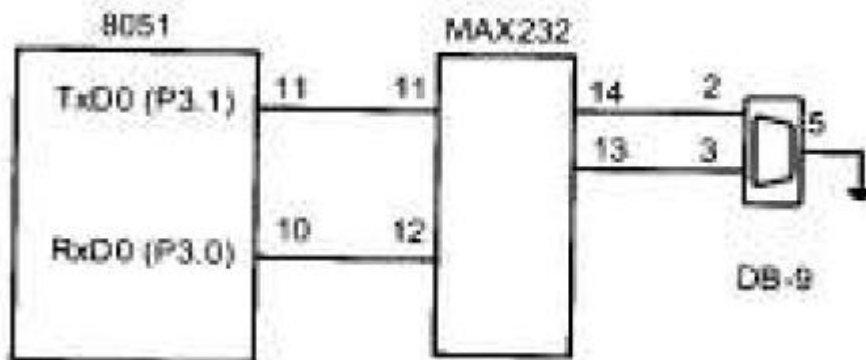
      JNB TF0, AGAIN    ; Monitor timer flag 0 until it rolls over

      CLR TR0          ; Stop timer 0

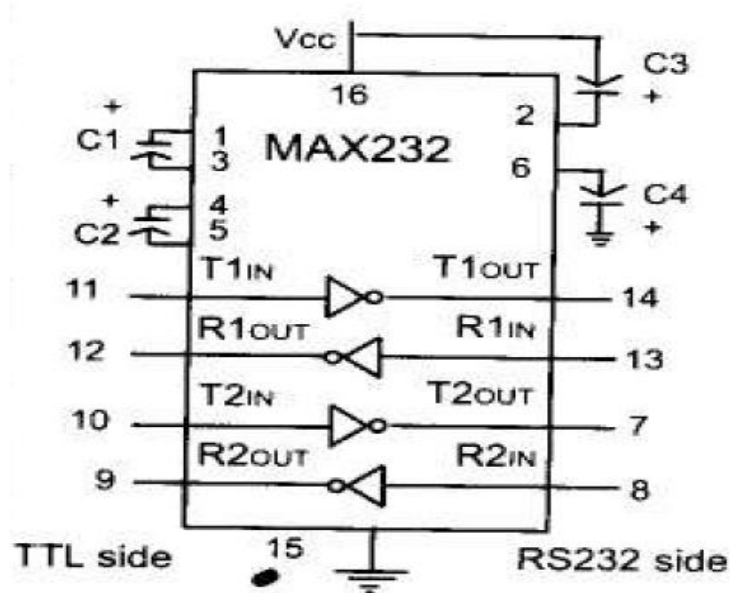
      CLR TF0          ; Clear timer 0 flag

      RET              ; Return to called program
```

6(b)



8051 has 2 pins namely TXD and RXD for transmitting and receiving serial data. These pins are TTL compatible. 8051 cannot be directly connected to RS232 due to difference in the voltage levels. Therefore a voltage converter is needed to convert RS 232 voltage levels to TTL voltage levels. A commonly used voltage converter is MAX 232. Fig below shows the pin diagram of MAX 232 and its connection to 8051.



- MAX232 has two set of line drivers for transferring and receiving data
- Line driver used for TxD are called T1 and T2, line drivers for RxD are called R1 and R2
- In MAX232 T1 line driver has designation of T1in and T1out on pin numbers 11 and 14 respectively
- T1in pin is TTL side and is connected to TxD (pin 11) of microcontroller, while T1out is RS232 side, that is connected to RxD (pin 2) of RS232 DB connector.
- R1 line driver has designation of R1in and R1out on pin numbers 13 and 12 respectively
- R1in (pin 13) is RS232 side and is connected to TxD (pin 3) of RS232 DB connector, while R1out (pin 12) is TTL side, that is connected to RxD (pin 10) of microcontroller.

7(a) Find the TH1 value needed to have the following baud rates.

(a) 9600

b) 4800

Assume XTAL = 11.0592 MHz

2.5 + 2.5 = 5

With XTAL = 11.0592 MHz, we have machine cycle frequency of 8051 = $11.0592/12 = 921.6$ KHz and $921.6\text{kHz}/32 = 28800$ Hz is the frequency provided by UART to timer 1 to set the baud rate.

(a) $28800 / 3 = 9600$ where -3 = FD(hex) is loaded into TH1

(b) $28800 / 6 = 4800$ where -6 = FA(hex) is loaded into TH1

7(b) Highest to lowest priority upon reset

1x5=5

External interrupt 0 (INT0)

Timer interrupt 0 (TF0)

External interrupt 1 (INT1)

Timer interrupt 1 (TF1)

Serial communication (RI + TI)

7(c) Programming of Serial communication Interrupt

(5 marks)

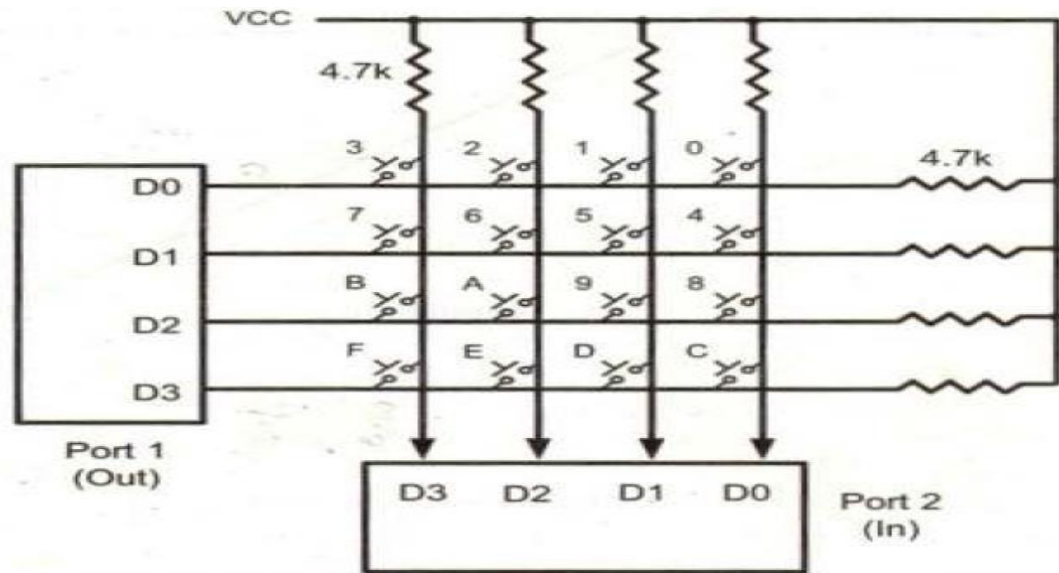
- In 8051 only one interrupt is set aside for serial communication
- This interrupt is used to both send and receive data
- If the interrupt bit in the IE register(IE.4) is enabled , when RI(received interrupt) or TI(transfer interrupt) is raised 8051 gets interrupted and jumps to memory address location 0023H to execute the ISR
- In ISR we must examine TI and RI flags to see which one caused the interrupt and respond accordingly
- In vast majority of applications, serial interrupt is used mainly for receiving the data and never used for sending data serially

SECTION-IV

8(a) Pin description of LCD

(5 marks)

Pin	symbol	Description
1	VSS	GROUND
2	VCC	+5 V POWER SUPPLY
3	VEE	POWER SUPPLY TO CONTROL CONTRAST
4	RS	RS=0 to select command register RS=1 to select data register
5	R/W	R/W=0 to write R/W=1 to read
6	E	Enable
7	DB0	The 8 bit data bus
8	DB1	The 8 bit data bus
9	DB2	The 8 bit data bus
10	DB3	The 8 bit data bus
11	DB4	The 8 bit data bus
12	DB5	The 8 bit data bus
13	DB6	The 8 bit data bus
14	DB7	The 8 bit data bus



8(c) C program to rotate stepper motor in clock wise direction

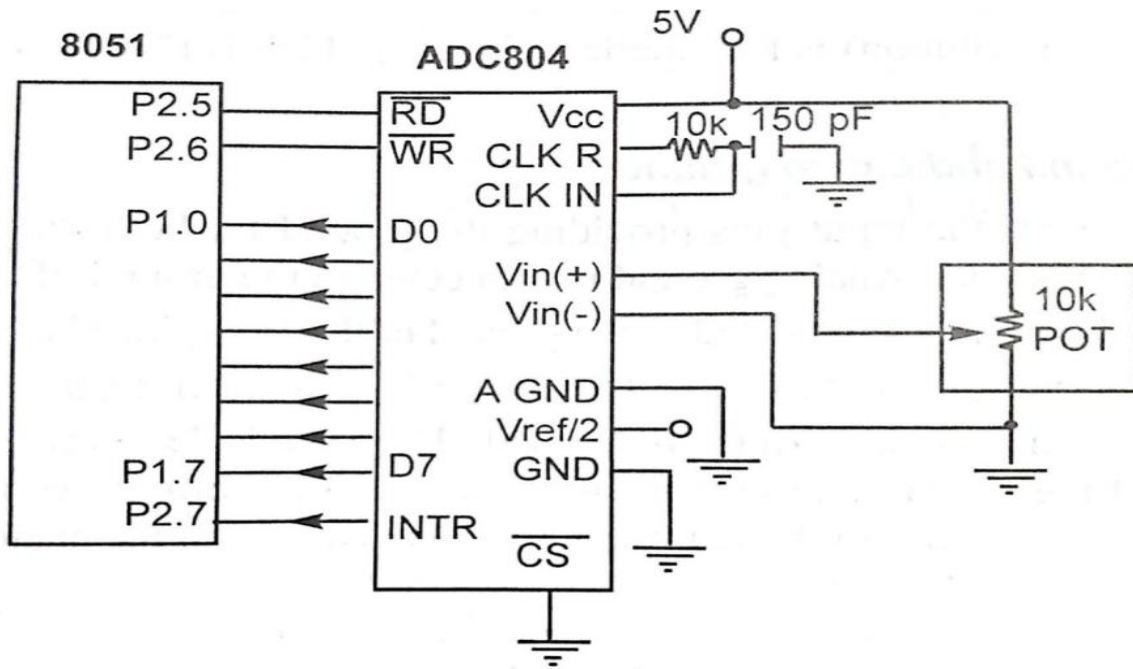
5

```

#include<reg51.h>
void delay( int value);
void main()
{
    while(1)
    {
        P1=0x66;
        delay(100);
        P1=0xcc;
        delay(100);
        P1=0x66;
        delay(100);
        P1=0x99;
        delay(100);
    }
}
void delay( int value)
{
    int i,j;
    for(i=0; i<value; i++)
    for(j=0; j<1275;j++);
}

```

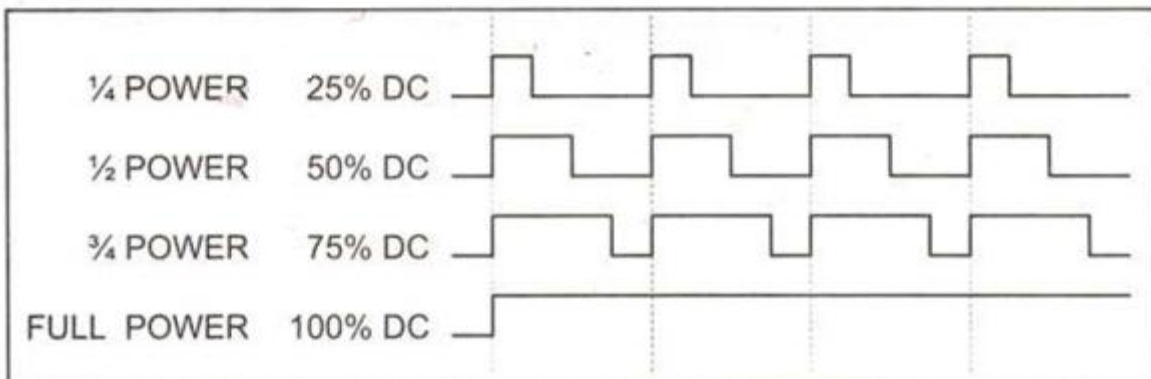
9(a) 8051 connection to ADC804 with self clock



9(b) Explain the concept of speed control of DC motor using PWM

5

The speed of the DC motor depends on three factors namely load, voltage and current. For a given load, the speed of the DC motor can be controlled by a method called pulse width modulation(PWM). In this method, the width of the pulse applied to the DC motor is changed causing increase or decrease in the power applied to the motor and thereby increasing or decreasing the motor speed. In other words when the duty cycle of the PWM is more the width of the pulse is large and hence the speed of the motor is high. As PWM control is widely used in the DC motor speed control some of the microcontrollers are available with PWM circuitry embedded in the chip. Fig below shows the PWM comparison for various powers.



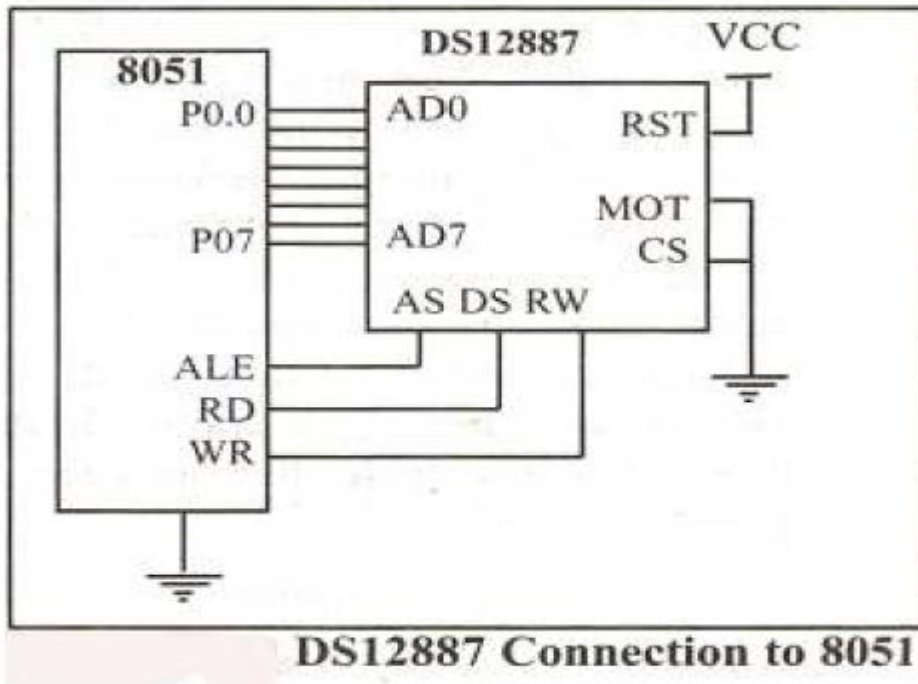
9(c) Write C program to interface temperature sensor to 8051

(5 marks)

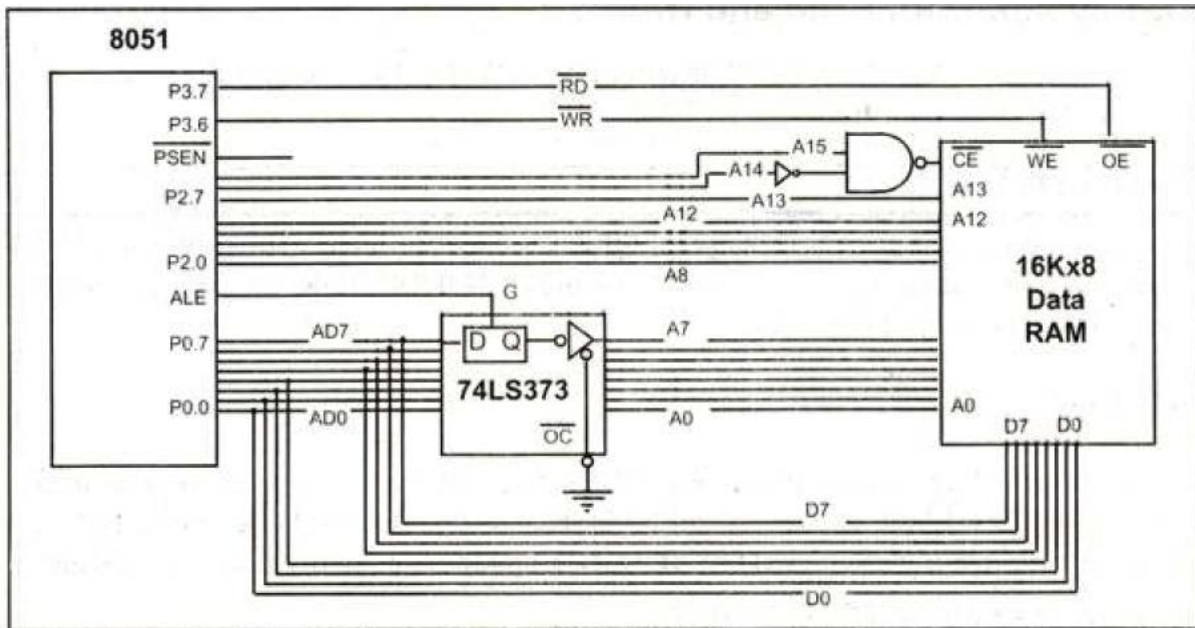
```
# include <reg51.h>
sbit RD = P2^5;
sbit WR = P2^6;
sbit INTR = P2^7;
sfr MYDATA = P1;
void main()
{
    unsigned char value;
    MYDATA = 0xFF;
    INTR = 1;
    RD = 1;
    WR = 1;
    while(1)
    {
        WR = 0;
        WR = 1;
        while ( INTR ==1);
        RD = 0;
        value = MYDATA;
        convert_and_display(value);
        RD =1;
    }
}
```

10 (a) Interface of 8051 to DS12887

(4 marks)



10 (b) 8051 connection to 16Kx 8 Data RAM (6 marks)



10(c) Relay connection to 8051

(5 marks)

Fig. shows 8051 connection to solid state relay. Solid state relay has no moving parts hence switching and response time is faster than electromechanical relay. It is widely used in controlling pumps, solenoids, alarms and other power applications. Fig. shows control of fan using solid state relay.

